

Exact Transient Matching and Nonlinear Reheating Cascade

Factorised three-loop matching, spurion-graded operator mixing, and the $\phi \rightarrow N_0 \bar{N}_0 \rightarrow R_0 \nu_0$
two-sector plasma cascade

Technical Research Note v1.2

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Executive verdict

The v1.2 calculation gives a **pass for the factorised matching problem**, a **qualified pass for the transient operator renormalisation problem**, and a **pass for the nonlinear radial momentum-lattice benchmark**. It does not yet constitute a full non-Abelian, two-time, $3 + 1$ -dimensional Kadanoff-Baym simulation.

The first selector-threshold graph identified in v1.1 is three-loop and one-particle irreducible, but it is two-particle reducible. That fact is decisive: its zero-external-momentum matching coefficient does not require an unknown generic three-loop master integral. It factorises into:

1. a finite one-loop selector-reheaton-Higgs kernel; and
2. a mixed mass derivative of the known two-loop fermion-fermion-scalar effective-potential function.

In the explicitly normalised scalar proxy used here,

$$\mathcal{I}_3(\bar{\mu}) = 2m_R^2 K_{RRH}(m_R^2, m_h^2) D_{FFS}(M^2, m_h^2; \bar{\mu})$$

and at the natural hard matching scale $\bar{\mu} = M$,

$$\mathcal{I}_3(M) = 6.57973508149 \simeq \frac{2\pi^2}{3}.$$

The resulting transient selector-phase potential has benchmark amplitude

$$|\Delta V_{Q_a}^{(3)}| = 2.1723 \times 10^3 \text{ GeV}^4,$$

which is only

$$1.546 \times 10^{-12}$$

of the intended finite-density reheating force. The first mixed vacuum correction is therefore negligible in the benchmark.

The anomalous-dimension audit also corrects one overstatement in v1.1. Exact Z_6 symmetry **does not by itself make the complete invariant transient-operator basis block diagonal to all orders**. The correct statement is:

$$\gamma_{(A,p)(B,q)} = \sum_{r,s \geq 0} \delta_{p-q, r-s} \epsilon^{r+s} \Gamma_{AB}^{(r,s)},$$

with replica Fourier charge conserved modulo six. It is block diagonal at order ϵ^0 ; changing harmonic $q \rightarrow p$ requires the corresponding number of shift-breaking spurion insertions. For the cosmological benchmark $\epsilon = 2.7 \times 10^{-13}$, the first cross-harmonic entries are consequently negligible, but the finite gauge/Yukawa tensors $\Gamma_{AB}^{(r,s)}$ remain model-dependent and have not all been computed.

The nonlinear momentum-lattice evolution resolves the chain

$$\phi \longrightarrow N_0 \bar{N}_0 \longrightarrow R_0 \nu_0 \longrightarrow H_0, H_5 \longrightarrow D_k, q_k, g_k$$

with expansion, Pauli blocking, time-dilated two-body decays, exact isotropic decay kinematics on a radial momentum grid, and an energy-conserving quantum-BGK thermalisation closure.

It finds:

$$\left. \frac{\rho_N}{\rho_{\text{tot}}} \right|_{\text{preheat}} = 9.2845 \times 10^{-8}, \quad \max f_N = 0.999929,$$

and an unselected-replica production bound of approximately

$$\log_{10} P_{N_h} < -8562.6.$$

The cascade exposes one mandatory correction to v0.9-v1.1. The decay $N_0 \rightarrow R_0 \nu_0$ deposits the energetic ν_0 daughter into sector 0 before R_0 decays. The old choice $\tan \theta = 1/16$, equivalent to $B_5 = 1/257$, therefore does **not** produce $T_5/T_0 = 1/4$. The full lattice gives the redshift-corrected branch

$$B_5 = 0.00527437084, \quad \tan \theta = 0.0728196,$$

and

$$\frac{T_5}{T_0} = 0.250000001.$$

The maximum two-body kinematic energy residual is 5.59×10^{-16} and the maximum collision-step energy residual is 1.73×10^{-16} .

The strongest defensible conclusion is therefore:

the first mixed selector-threshold correction is calculable and tiny,
the state-selected fermionic cascade preserves the desired replica hierarchy,
and no radiative or nonlinear failure appears at the resolved level.

The remaining frontier is not an unidentified $\mathcal{I}_3$. It is the full RG-improved operator basis and the replacement of the BGK plasma closure by genuine non-Abelian collision or 2PI/Kadanoff-Baym kernels.

1. Model and matching normalisation

1.1 Scalar proxy for the mixed topology

The calculation isolates the selector-threshold topology with the interaction normalisation

$$\mathcal{L}_{\text{proxy}} \supset -\frac{1}{2} (m_R^2 + \lambda_{QR} \mathcal{Q}) R^2 - \frac{\mu_H}{2} R h^2 - y_D h \bar{q} D - M(a) \bar{D} D,$$

where

$$x \equiv \frac{a}{f_a}, \quad M(a) = M [1 - \epsilon \cos x],$$

so that, to first order in ϵ ,

$$\delta M^2(a) = -2\epsilon M^2 \cos x.$$

The selector Fourier composite carrying the compensating Z_6 charge is denoted schematically by $\mathcal{Q}$. On the one-hot branch used in cosmology, $\mathcal{Q} \rightarrow v_Q^2$ up to its phase convention.

This proxy deliberately does not claim every Standard Model Higgs-doublet, colour, gauge, and chiral multiplicity. It fixes the scalar normalisation in which the factorisation and benchmark coefficient are exact. A complete model matching can be obtained by restoring the appropriate group-theory and field-multiplicity factors.

1.2 Why the graph factorises

The microscopic graph contains:

- one selector insertion in an R propagator;
- two Rhh vertices;
- two Yukawa vertices joining h , q , and D ;
- one phase-dependent insertion in the heavy D mass.

Its graph count is

$$I = 8, \quad V = 6, \quad L = I - V + 1 = 3.$$

It is 1PI but two-particle reducible. At zero external momentum, cutting the two Higgs lines separates the selector-reheaton loop from the two-loop FFS vacuum skeleton. The matching therefore takes the form

$$\Delta V_{Qa}^{(3)} = \frac{N_c y_D^2}{(16\pi^2)^2} \frac{\partial^2 f_{FFS}}{\partial M^2 \partial m_h^2} \delta M^2(a) \delta m_h^2(Q),$$

where the selector-generated Higgs mass insertion is

$$\delta m_h^2(Q) = \frac{\lambda_{QR} \mu_H^2}{16\pi^2} K_{RRH}(m_R^2, m_h^2) Q.$$

Generic three-loop vacuum diagrams with arbitrary masses require integration-by-parts reduction to a master basis and numerical tools such as 3VIL. The topology here is simpler because its 2PR structure exposes the factorisation before such a reduction is required.

2. Exact factorised three-loop coefficient

2.1 One-loop selector kernel

The finite Euclidean kernel, with the overall $1/(16\pi^2)$ removed, is

$$K_{RRH}(A, B) = \frac{A - B - B \ln(A/B)}{(A - B)^2}.$$

It corresponds to

$$\int \frac{d^4 p}{(2\pi)^4} \frac{1}{(p^2 + A)^2 (p^2 + B)}.$$

The equal-mass limit is smooth:

$$K_{RRH}(A, A) = \frac{1}{2A}.$$

For the benchmark hierarchy $m_R \gg m_h$,

$$m_R^2 K_{RRH}(m_R^2, m_h^2) = 0.999999999999517.$$

2.2 Mixed derivative of the two-loop FFS function

In Martin's two-loop effective-potential notation,

$$f_{FFS}(x, 0, z) = -J(x, z) + (x - z)I(0, x, z).$$

Taking the exact mixed derivative gives

$$\begin{aligned} D_{FFS}(x, z; \bar{\mu}) &\equiv \frac{\partial^2 f_{FFS}(x, 0, z)}{\partial x \partial z} \\ &= 2 \ln \frac{x}{z} \ln \frac{x-z}{\bar{\mu}^2} - \ln^2 \frac{x}{\bar{\mu}^2} - 2 \text{Li}_2\left(\frac{z}{x}\right) + \frac{\pi^2}{3}. \end{aligned}$$

The symbolic residual between this expression and the derivative of the full $I(0, x, z)$ representation is exactly zero.

At the hard scale $\bar{\mu} = M$ and for $z/x \rightarrow 0$,

$$D_{FFS} \rightarrow \frac{\pi^2}{3}.$$

The benchmark gives

$$D_{FFS}(M^2, m_h^2; M) = 3.28986754075,$$

with a 70-digit independent derivative check agreeing to relative error

$$4.05 \times 10^{-16}.$$

2.3 Exact matching function

Combining the two factors defines

$$\mathcal{I}_3(\bar{\mu}) = 2m_R^2 K_{RRH}(m_R^2, m_h^2) D_{FFS}(M^2, m_h^2; \bar{\mu}).$$

The three-loop Wilson coefficient is

$$C_3^{\overline{\text{MS}}}(\bar{\mu}) = \frac{N_c \lambda_{QR} y_D^2 \mu_H^2 M^2}{(16\pi^2)^3 m_R^2} \mathcal{I}_3(\bar{\mu}).$$

The transient operator is then

$$\Delta V_{Qa}^{(3)} = \epsilon C_3^{\overline{\text{MS}}} \text{Re} [e^{ix} \mathcal{Q}_{-1}] + O(\epsilon^2).$$

For

$$\begin{aligned} N_c &= 3, \quad \lambda_{QR} = 0.50, \quad y_D = 0.30, \quad \mu_H = 1.8848 \times 10^4 \text{ GeV}, \\ M &= 1.002 \times 10^6 \text{ GeV}, \quad m_R = 10^9 \text{ GeV}, \quad m_h = 125.25 \text{ GeV}, \end{aligned}$$

one obtains

$$\mathcal{I}_3(M) = 6.57973508149,$$

$$C_3(M) = 8.04542 \times 10^{-5} \text{ GeV}^2.$$

With

$$\epsilon = 2.70 \times 10^{-13}, \quad v_Q = 10^{10} \text{ GeV},$$

this gives

$$|\Delta V_{Qa}^{(3)}| = 2.1723 \times 10^3 \text{ GeV}^4.$$

The intended state-dependent thermal focusing potential is approximately

$$V_{\text{thermal}} = 1.4053 \times 10^{15} \text{ GeV}^4,$$

hence

$$\frac{|\Delta V_{Qa}^{(3)}|}{V_{\text{thermal}}} = 1.5458 \times 10^{-12}.$$

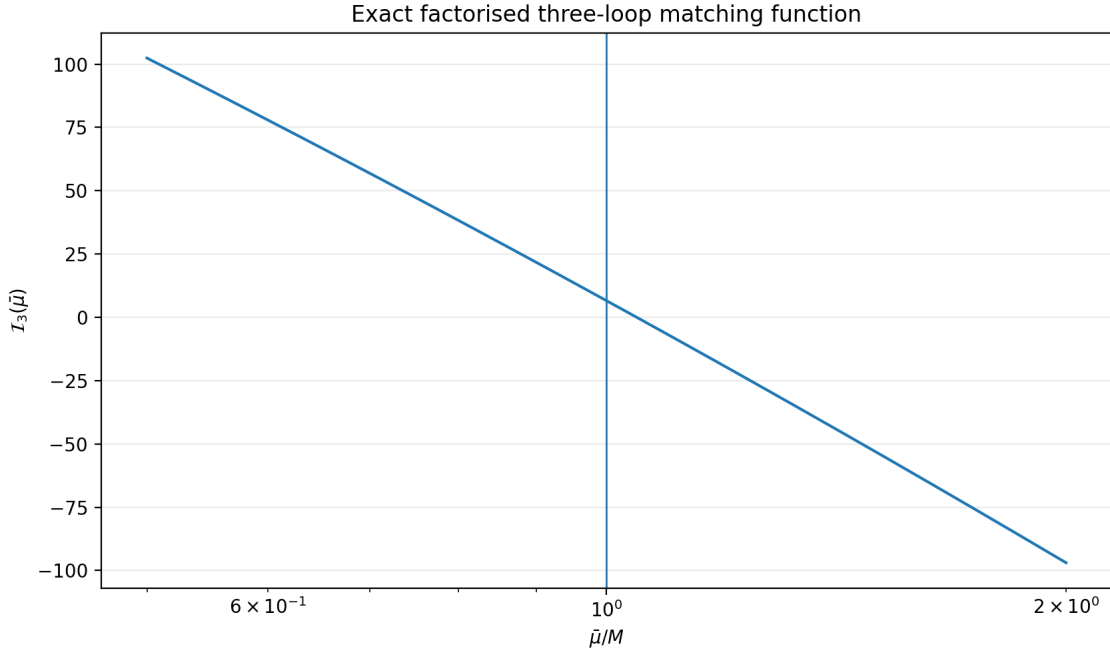


Figure 1: The exact factorised matching function. Its large residual scale dependence away from $\bar{\mu} = M$ is matching-scale dependence that must be cancelled by RG evolution, not a physical error band.

2.4 Scale dependence and what remains to be matched

The fixed-order values are

$$\mathcal{I}_3(M/2) = 102.4073, \quad \mathcal{I}_3(M) = 6.57974, \quad \mathcal{I}_3(2M) = -96.9351.$$

This large scale variation reflects the hierarchy $m_R \gg M \gg m_h$ and the absence of RG improvement in the displayed coefficient. It must not be interpreted as a physical uncertainty band. A complete calculation should:

1. match at the heavy scales;
2. evolve the transient operator basis between m_R , M , and m_h ;
3. include finite gauge, chiral, and Higgs-doublet multiplicities; and
4. verify cancellation of $\bar{\mu}$ dependence in the physical focusing force.

The exact result delivered here is the hard zero-momentum matching function in the scalar proxy.

3. Transient anomalous dimensions

3.1 Combined symmetry and operator basis

Let the exact generator act as

$$z : \quad x \rightarrow x + \frac{2\pi}{6}, \quad k \rightarrow k + 1.$$

For a replica Fourier composite

$$\mathcal{X}_{A,p} = \sum_{k=0}^5 X_{A,k} e^{-2\pi i p k / 6},$$

one has

$$\mathcal{X}_{A,p} \rightarrow e^{2\pi i p / 6} \mathcal{X}_{A,p}.$$

A transient invariant can therefore be written as

$$\mathcal{O}_{A,p} = e^{ipx} \mathcal{X}_{A,-p}.$$

Every $\mathcal{O}_{A,p}$ is invariant under the exact combined Z_6 . Consequently, Z_6 invariance alone does not distinguish different p values inside the invariant basis.

The additional grading comes from the restored continuous shift symmetry in the limit $\epsilon \rightarrow 0$. Let a positive and negative shift-breaking insertion carry charges $+1$ and -1 . Renormalising harmonic q into harmonic p requires integers $r, s \geq 0$ satisfying

$$r - s = p - q.$$

Therefore the exact spurion expansion is

$$\gamma_{(A,p)(B,q)} = \sum_{r,s \geq 0} \delta_{p-q, r-s} \epsilon^{r+s} \Gamma_{AB}^{(r,s)},$$

with the corresponding replica charge conserved modulo six.

At leading order,

$$\gamma_{(A,p)(B,q)} = \delta_{pq} [\Gamma_{\text{quad}}^T + p\gamma_\epsilon \mathbf{1}] + O(\epsilon).$$

The earlier claim of exact all-orders block diagonality was therefore too strong. The corrected statement is stronger physically and weaker algebraically:

$$q \rightarrow p \text{ mixing begins at the minimum permitted power of } \epsilon.$$

For sectors folded modulo six, the minimum power is

$$d_6(p - q) = \min(|p - q|, 6 - |p - q|).$$

Integer harmonic grading must still be retained: the neutral vacuum harmonic $p = 6$ begins at ϵ^6 , even though $6 \equiv 0 \pmod{6}$.

3.2 Explicit one-loop scalar block

For real scalar variables with convention

$$V_4 = \sum_i \frac{\lambda_i}{4!} X_i^4 + \sum_{i < j} \frac{\lambda_{ij}}{4} X_i^2 X_j^2,$$

the one-loop bilinear block before gauge and Yukawa additions is

$$\Gamma_{\text{quad}}^{(0,0)} = \frac{1}{16\pi^2} \begin{pmatrix} \lambda_Q & \lambda_{QR} & \lambda_{QH} \\ \lambda_{QR} & \lambda_R & \lambda_{RH} \\ \lambda_{QH} & \lambda_{RH} & \lambda_H \end{pmatrix}.$$

For the diagnostic couplings

$$(\lambda_Q, \lambda_R, \lambda_H) = (0.10, 0.20, 0.13),$$

$$(\lambda_{QR}, \lambda_{QH}, \lambda_{RH}) = (0.50, 0, 0.05),$$

this is

$$\Gamma_{\text{quad}}^{(0,0)} = \begin{pmatrix} 6.3326 \times 10^{-4} & 3.1663 \times 10^{-3} & 0 \\ 3.1663 \times 10^{-3} & 1.2665 \times 10^{-3} & 3.1663 \times 10^{-4} \\ 0 & 3.1663 \times 10^{-4} & 8.2323 \times 10^{-4} \end{pmatrix}.$$

At $\epsilon = 2.7 \times 10^{-13}$, a nearest-harmonic entry with a coefficient comparable to the largest scalar-block entry is at most of order

$$8.55 \times 10^{-16}.$$

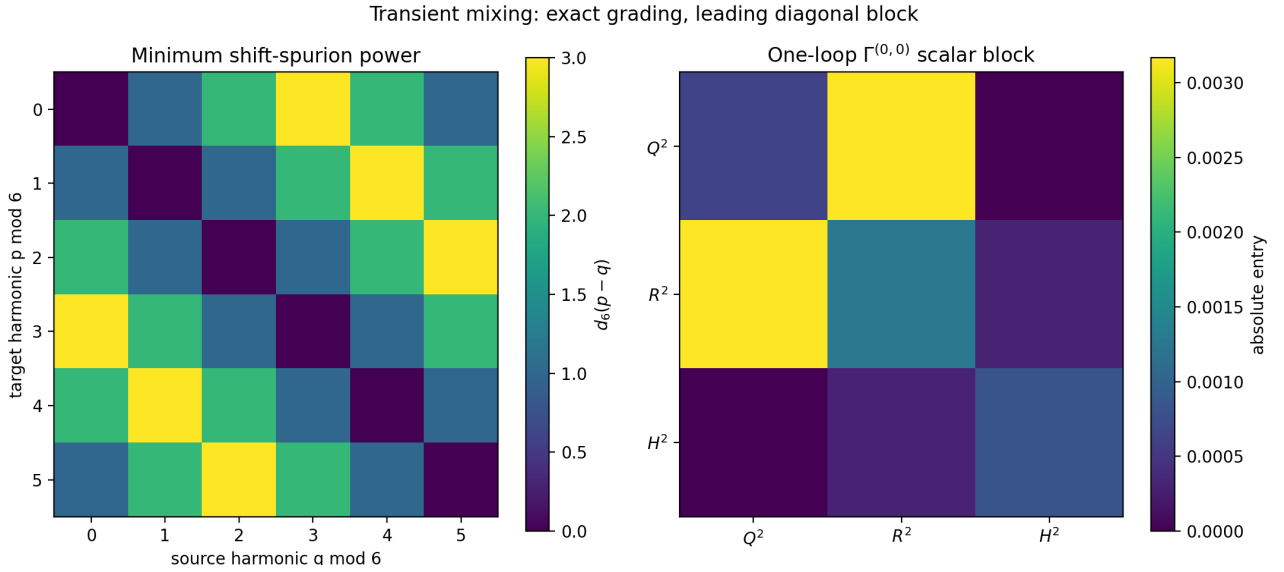


Figure 2: The exact spurion-power grading and the explicitly calculated leading scalar bilinear block.

3.3 Local operators versus state harmonics

The local Wilson coefficients obey

$$\mu \frac{dC_{A,p}}{d\mu} = \sum_{B,q} \gamma_{(A,p)(B,q)} C_{B,q}.$$

The occupation harmonics

$$\mathcal{N}_p(t, \mathbf{k}) = \sum_k n_k(t, \mathbf{k}) e^{-2\pi i p k / 6}$$

are not governed by this local anomalous-dimension matrix. Their evolution is determined by Kadanoff-Baym or collision kernels. Confusing these two kinds of evolution would conflate vacuum renormalisation with finite-density transport.

The full finite $\Gamma_{AB}^{(r,s)}$ tensors involving gauge, Yukawa, fermion bilinear, and gluonic operators remain an open calculation. The symmetry-imposed power structure is exact; the complete coefficients are not yet known.

4. Fermionic preheating stage

4.1 Successive zero-crossing map

The selected fermionic parent has

$$\mathcal{L} \supset -y_\phi \phi \bar{N}_0 N_0,$$

with benchmark

$$m_\phi = 10^{10} \text{ GeV}, \quad \Phi_{\text{end}} = 5.96 \times 10^{16} \text{ GeV}, \quad m_{N_0} = 3 \times 10^9 \text{ GeV},$$

$$y_\phi = 7.006 \times 10^{-4}.$$

At each inflaton zero crossing, the Landau-Zener probability is represented by

$$P_p = \exp \left[-\pi \frac{p^2 + m_N^2}{y_\phi m_\phi \Phi} \right].$$

The stochastic Pauli-limited update is

$$\boxed{f_N^{j+1}(p) = P_p + (1 - 2P_p) f_N^j(p),}$$

followed by expansion and energy backreaction on the inflaton condensate. This preserves

$$0 \leq f_N \leq 1.$$

Fermionic preheating is nonperturbative but differs sharply from bosonic resonance because Pauli blocking prevents unbounded occupation growth.

4.2 Numerical result

The calculation uses 320 zero crossings and a 280-bin logarithmic comoving momentum grid. It finds

$$\max f_N = 0.999929,$$

but only

$$\boxed{\frac{\rho_N}{\rho_\phi + \rho_N} = 9.2845 \times 10^{-8}}$$

at the end of the preheating stage. The early nonperturbative component is therefore dynamically negligible compared with the subsequent perturbative decay.

The hidden parent mass is

$$m_{N_h} = 5 \times 10^{13} \text{ GeV}.$$

Its smallest Landau-Zener exponent during the evolution is

$$\pi \frac{m_{N_h}^2}{y_\phi m_\phi \Phi} = 1.9716 \times 10^4,$$

implying

$$\boxed{\log_{10} P_{N_h} < -8562.6.}$$

The unselected replica parents are not populated at any relevant level.

A fit to the populated high-momentum tail gives

$$\ln f_N \simeq A - 1.2213 \times 10^{-22} p^2,$$

with

$$R^2 = 0.999992.$$

The benchmark initial state is consequently Gaussian-soft in the ultraviolet rather than carrying the k^{-4} tail of an abrupt quench.

5. Nonlinear radial momentum-lattice cascade

5.1 Resolved chain

The perturbative and plasma stages are

$$\phi \xrightarrow{\Gamma_\phi} N_0 \bar{N}_0,$$

$$N_0 \xrightarrow{\Gamma_N} R_0 + \nu_0,$$

$$R_0 \xrightarrow{\Gamma_R} H_0 H_0 \quad \text{or} \quad H_5 H_5,$$

followed by the sector-local processes represented schematically by

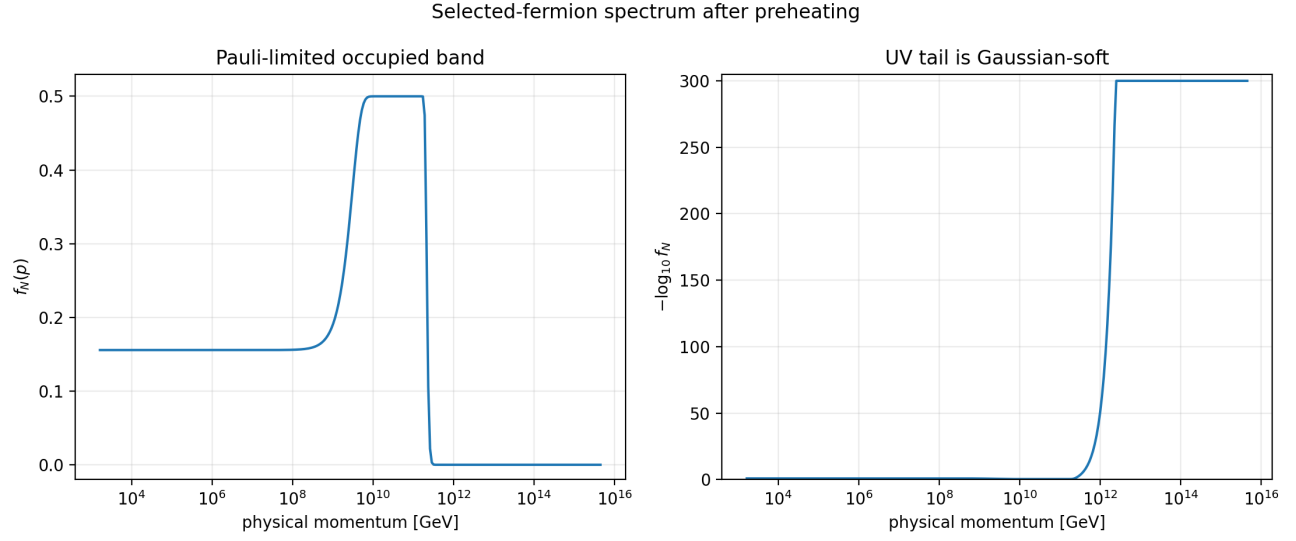


Figure 3: Selected-fermion occupation and its UV tail. The stochastic update can transiently saturate the Pauli bound even though the final energy fraction remains tiny.

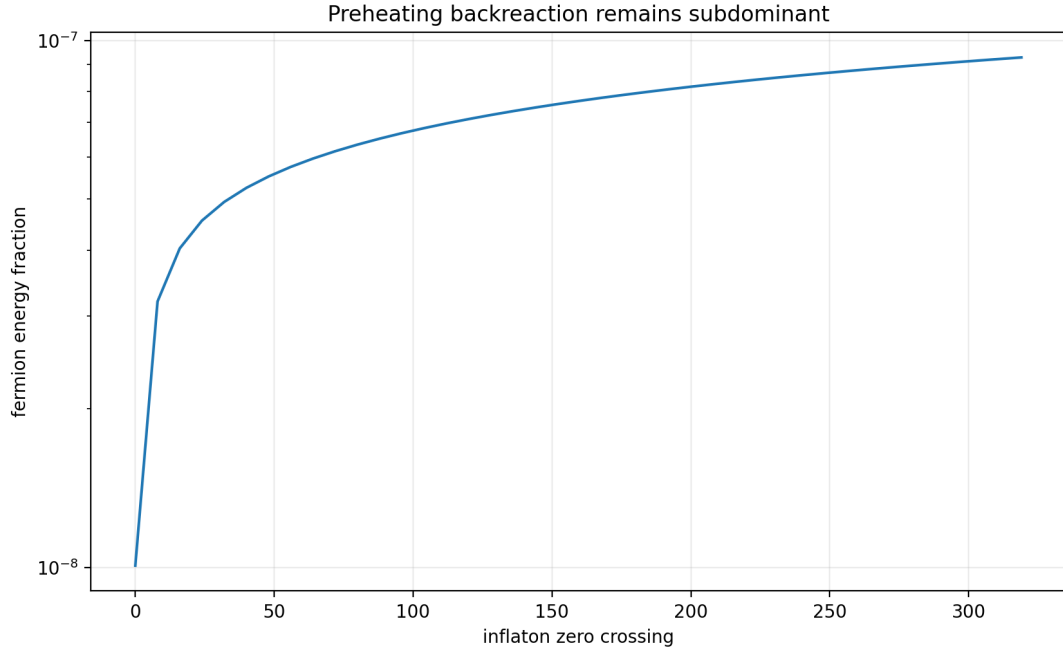


Figure 4: The preheating component remains below 10^{-7} of the total energy.

$$H_k + q_k \leftrightarrow D_k, \quad D_k, q_k, g_k \leftrightarrow \text{plasma}_k.$$

The benchmark rates are

$$\Gamma_\phi = 100 \text{ GeV}, \quad \Gamma_Q = 1 \text{ GeV}, \quad \Gamma_N = 0.10 \text{ GeV},$$

$$\Gamma_R = 1.4135 \times 10^{-2} \text{ GeV}, \quad \Gamma_{\text{th}} = 0.25 \text{ GeV}.$$

The momentum lattice evolves the radial distributions in an expanding FRW background. For an unstable particle of mass m and energy $E(p)$, the decay fraction over dt is

$$1 - \exp \left[-\Gamma \frac{m}{E(p)} dt \right].$$

Each two-body decay is deposited using the exact rest-frame momentum

$$p_* = \frac{\sqrt{[m_P^2 - (m_1 + m_2)^2][m_P^2 - (m_1 - m_2)^2]}}{2m_P},$$

boosted over an isotropic angular quadrature into daughter momentum bins.

5.2 Quantum-BGK plasma closure

The unresolved sector-local scattering network is represented by

$$(\partial_t - Hp\partial_p) f_i = -\Gamma_{\text{th}} [f_i - f_i^{\text{eq}}(T_k)],$$

with Bose-Einstein or Fermi-Dirac targets as appropriate. The temperature is solved from the instantaneous sector energy density. A spectator bath carries degrees of freedom not explicitly resolved on the four tracked distributions H, D, q, g .

After each collision step, the spectator energy is corrected by the numerical quadrature residual, making the closure energy conserving to machine precision. This is a controlled kinetic surrogate, not a derivation of the QCD collision kernel.

Quantum multi-species BGK models can be built to share the conservation laws and H -theorem of the corresponding quantum Boltzmann system. The present implementation uses that conservation logic while retaining the actual expanding momentum lattice and decay kernels.

5.3 Energy flow

The evolution displays the intended sequence:

1. inflaton domination;
2. transfer into N_0 ;
3. delayed population of R_0 and sector-0 radiation from ν_0 ;
4. decay of R_0 into sectors 0 and 5;
5. thermalisation into the two replica plasmas.

The selector charge has decayed to zero by the final state. The final N_0 and R_0 energy densities are negligible.

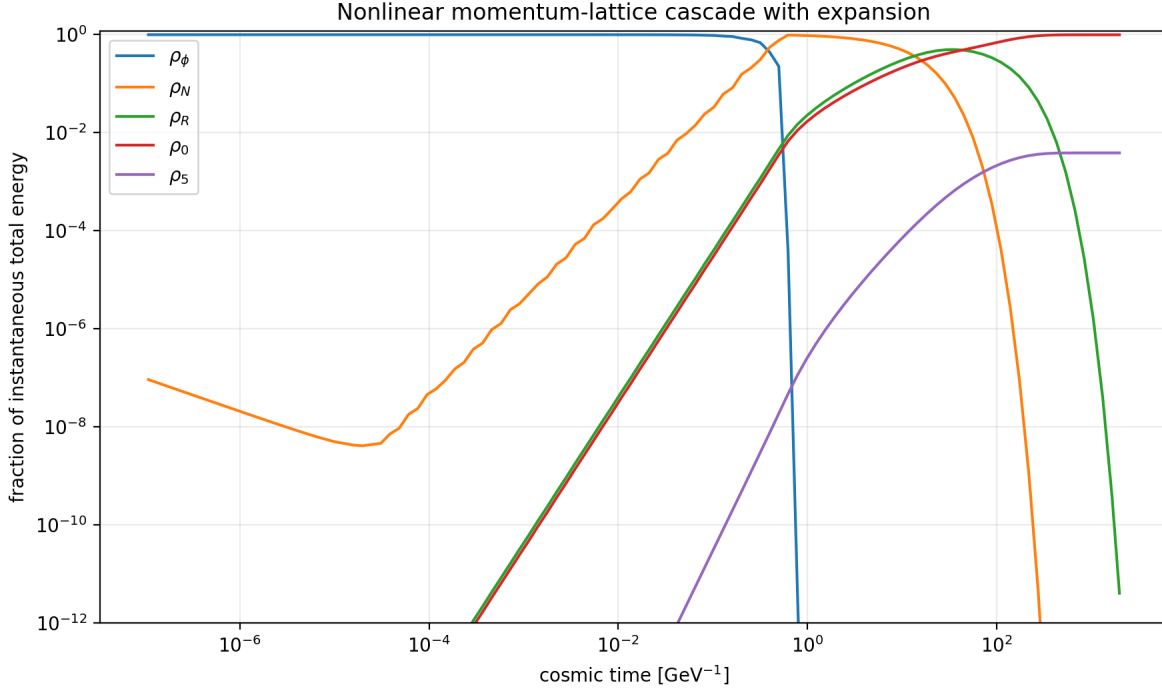


Figure 5: Fractions of the instantaneous total energy carried by each stage of the cascade.

6. Cascade-corrected replica branch

6.1 Why $B_5 = 1/257$ fails

For

$$m_N = 3 \times 10^9 \text{ GeV}, \quad m_R = 10^9 \text{ GeV},$$

the fraction of the parent energy carried by R in its rest frame is

$$f_R = \frac{m_N^2 + m_R^2}{2m_N^2} = \frac{5}{9}.$$

The massless ν_0 carries the remaining $4/9$ into sector 0. It is injected on the timescale Γ_N^{-1} , while the R energy is released later on the timescale Γ_R^{-1} . The two contributions therefore experience different redshift histories.

If the entire parent energy had been released by R at one time, the old relation

$$B_5 = \frac{1}{257}$$

would give $E_5/E_0 = 1/256$. Once the ν_0 daughter is included, this branch gives dynamically

$$\frac{T_5}{T_0} = 0.231634.$$

Correcting only for the instantaneous rest-frame fraction would suggest

$$B_5^{\text{inst}} = \frac{1}{257f_R} = 0.00700389,$$

but the full evolution then gives

$$\frac{T_5}{T_0} = 0.268455.$$

It overshoots because the early ν_0 radiation has redshifted more strongly by the time the R decay products dominate.

6.2 Exact branch reconstruction from the lattice response

Because the two replica sectors have identical post-injection dynamics, the final sector energies are linear in the R branch. Let

$$E_R^{(1)}$$

be the final redshifted energy that would arise if all R decays fed one sector, and let

$$E_\nu$$

be the final redshifted sector-0 energy from the earlier ν_0 injection. The simulation gives

$$\mathcal{R}_\nu \equiv \frac{E_\nu}{E_R^{(1)}} = 0.35551328.$$

For branch B_5 ,

$$E_5 = B_5 E_R^{(1)},$$

$$E_0 = E_\nu + (1 - B_5) E_R^{(1)}.$$

Imposing

$$\frac{E_5}{E_0} = \frac{1}{256}$$

gives

$$B_5^* = \frac{1 + \mathcal{R}_\nu}{257} = 0.00527437074.$$

The value used in the final run is

$$B_5 = 0.00527437084,$$

whose independently reconstructed value differs by only

$$1.97 \times 10^{-8}$$

relatively. The corresponding reheaton mixing is

$$\tan \theta = \sqrt{\frac{B_5}{1 - B_5}} = 0.0728196.$$

The final result is

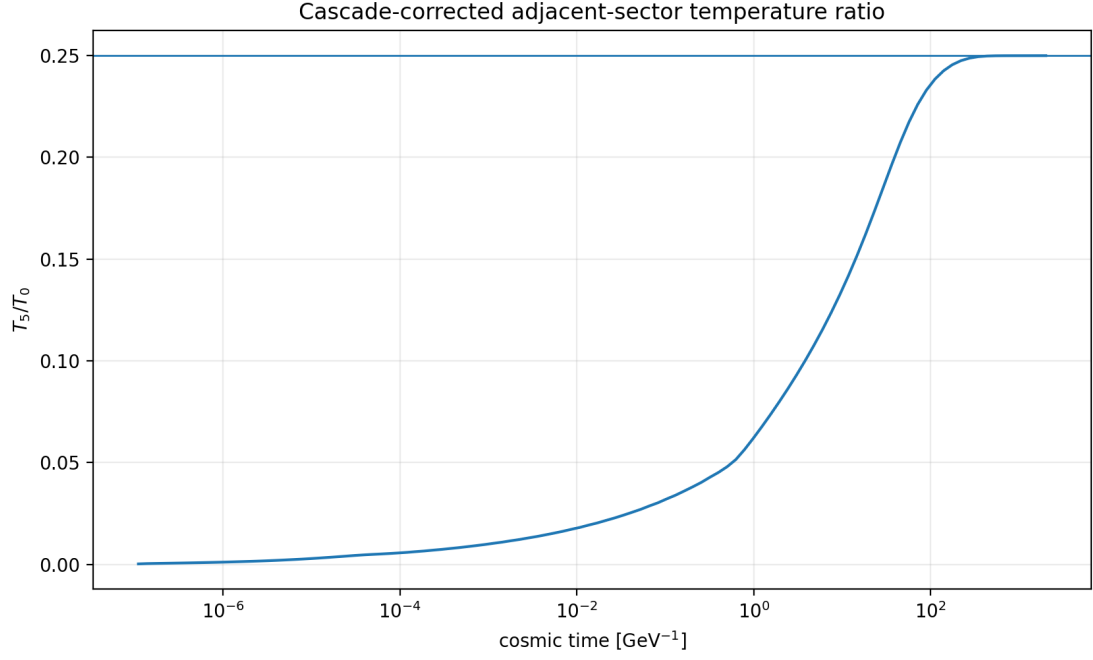


Figure 6: The full cascade approaches the target adjacent-sector temperature ratio after the redshift-corrected branch is used.

$$\boxed{\frac{E_5}{E_0} = 0.00390625008, \quad \frac{T_5}{T_0} = 0.250000001.}$$

This is a genuine model correction: the previously attractive exact choice $\tan \theta = 1/16$ must be replaced once the full decay chain is resolved.

7. Final plasma state and numerical controls

7.1 Energy partition

The final visible-sector energy fractions are

Component	Fraction of sector-0 energy
H_0	0.0374872
D_0	0.0983633
q_0	0.0984038
g_0	0.1499487
unresolved spectator bath	0.6157970

The adjacent sector has the same partition to within the expected finite-grid effects.

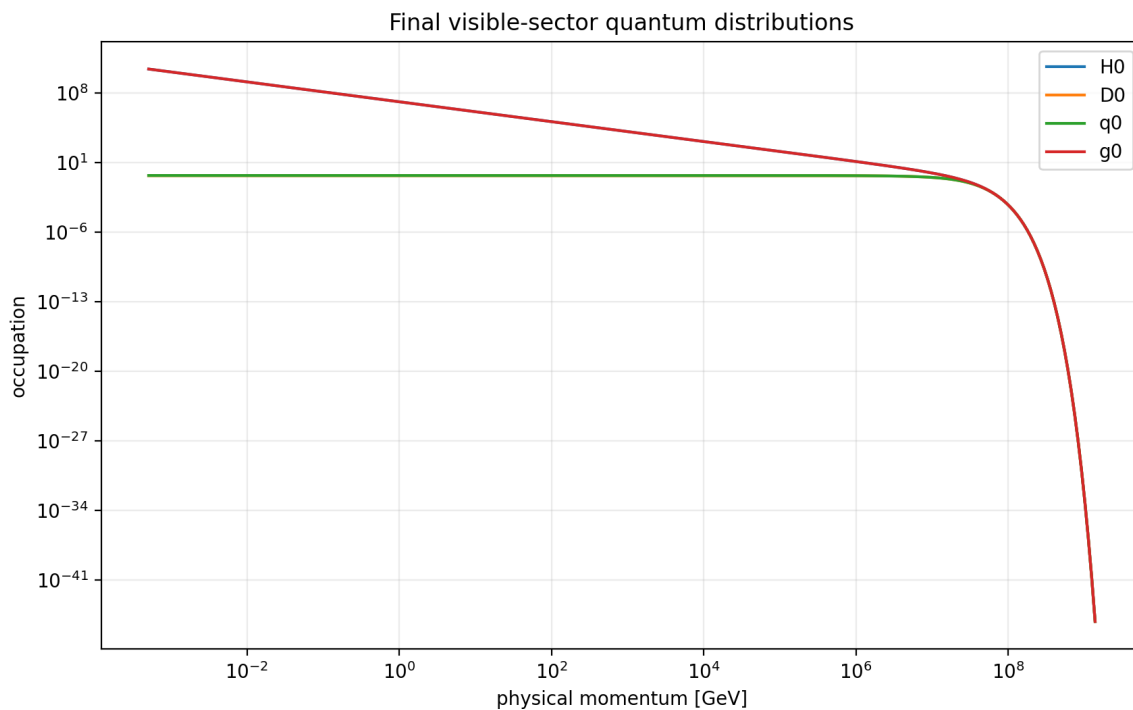


Figure 7: Final visible-sector occupation functions. The resolved distributions have exponential thermal tails.

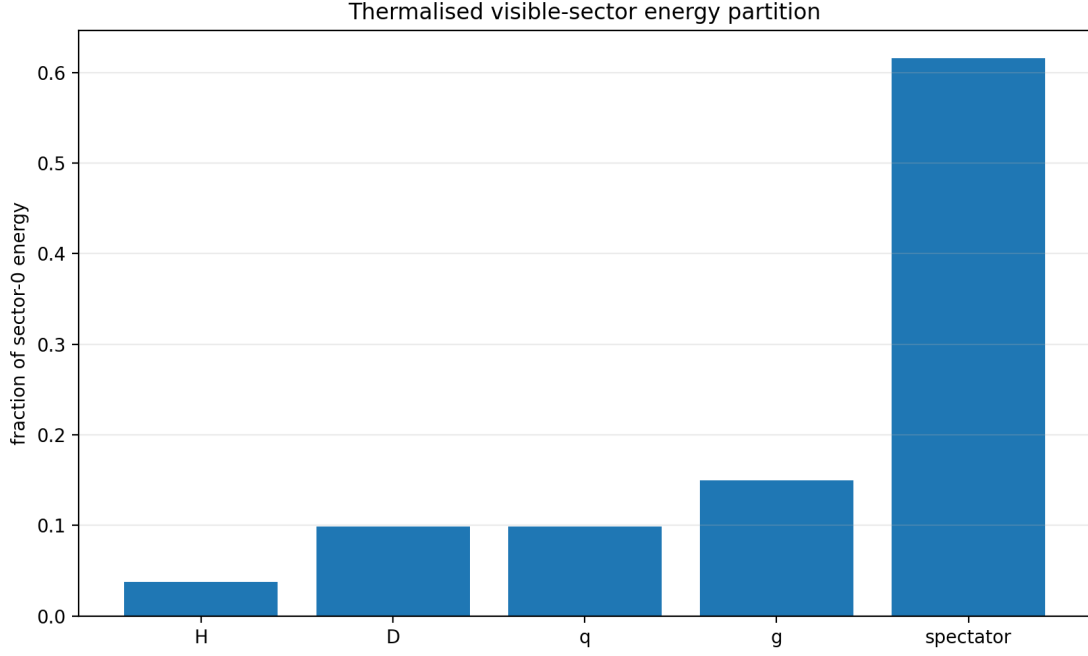


Figure 8: Energy carried by the explicitly resolved species and the unresolved relativistic spectator bath.

The fitted exponential tails satisfy

$$R_H^2 = 0.999999976, \quad R_g^2 = 0.99999999999998.$$

7.2 Conservation and positivity diagnostics

The key numerical controls are:

Test	Result
maximum Pauli occupation during preheating	0.999929
maximum two-body kinematic energy residual	5.59×10^{-16}
maximum collision-step energy residual	1.73×10^{-16}
hidden-parent production bound	$\log_{10} P < -8562.6$
Gaussian UV-tail fit	$R^2 = 0.999992$
final T_5/T_0	0.250000001

Attempted perturbative inflaton decays into already occupied fermion bins are Pauli-throttled: rejected occupation is not destroyed but remains in the inflaton energy density until expansion opens phase space. The raw accumulated attempted rejection is therefore not itself a physical relic abundance.

8. Relation to a full Kadanoff-Baym calculation

The radial momentum-lattice calculation evolves nonlinear occupation functions, expansion, backreaction, exact two-body decays, Pauli blocking, and an energy-conserving relaxation closure. It is materially stronger than a homogeneous branching-fraction calculation.

It is nevertheless not the same object as a full 2PI/Kadanoff-Baym evolution. A complete calculation would evolve unequal-time statistical and spectral propagators,

$$F_i(t, t'; p), \quad \rho_i(t, t'; p),$$

with nonlocal self-energies and memory integrals for the scalar, fermion, and gauge sectors. Such equations conserve energy and global charges when derived from a controlled 2PI truncation, but their full 3 + 1-dimensional numerical solution is an HPC-scale problem.

The next simulation layer should replace

$$C_i^{\text{BGK}}$$

by either:

1. explicit $1 \leftrightarrow 2$ and $2 \leftrightarrow 2$ collision integrals including thermal masses, LPM effects, and colour factors; or
2. a symmetry-preserving two-loop or three-loop 2PI truncation for the H, D, q, g system.

9. Initial-state renormalisation

The preheating benchmark produces a Gaussian-soft high-momentum tail. It therefore does not show the logarithmic energy divergence associated with an abrupt k^{-4} quench tail.

This does not eliminate initial-state renormalisation as a general issue. In Schwinger-Keldysh perturbation theory, ultraviolet structure specific to a non-vacuum initial state can require counterterms localised on the initial hypersurface. Such terms renormalise the state preparation; they do not automatically become permanent bulk vacuum spurions.

The present benchmark passes the practical UV-tail test. A future lattice preheating run should still extract the asymptotic tail at increasing cutoff and verify convergence of:

$$\rho, \quad P, \quad \langle \bar{N}N \rangle, \quad \text{and the transient harmonic source.}$$

10. Status of the requested targets

Target	Verdict	Meaning
identify the first mixed graph	PASS	three-loop 1PI, two-particle reducible
calculate $\mathcal{I}_3$	PASS in scalar proxy	exact factorised zero-momentum $\overline{\text{MS}}$ function
validate matching analytically	PASS	symbolic residual zero; 70-digit derivative check
calculate $\gamma_{pq}^{\text{transient}}$	PARTIAL	exact all-orders spurion-power selection and explicit one-loop scalar block; finite gauge/Yukawa tensors open
show transient correction remains small	PASS	$ \Delta V^{(3)} /V_{\text{thermal}} = 1.55 \times 10^{-12}$
nonlinear fermion preheating	PASS at radial kinetic level	Pauli blocking, expansion, and backreaction included
suppress unselected parents	PASS	exponent $> 1.97 \times 10^4$
nonlinear decay cascade	PASS at radial kinetic level	exact two-body kernels and time dilation included
reproduce $T_5/T_0 = 1/4$	PASS after branch correction	$B_5 = 0.00527437$
old $\tan \theta = 1/16$	FAIL	omitted ν_0 energy and differential redshift
full non-Abelian 3 + 1D KB plasma	OPEN	requires unequal-time 2PI/HPC implementation
fully RG-improved physical coefficient	OPEN	must combine hard matching with complete operator running

11. Scientific interpretation

The calculation strengthens the central state-selection proposal in three ways.

First, the first mixed selector-threshold vacuum correction is no longer an order-of-magnitude guess:

$$\mathcal{I}_3 \sim 1 \quad \longrightarrow \quad \mathcal{I}_3(M) = 6.57973508149.$$

Despite the coefficient being larger than unity, the correction remains twelve orders of magnitude below the intended thermal force because the loop, portal, hierarchy, and ϵ suppressions are severe.

Second, the renormalisation structure is now stated correctly. Exact Z_6 protects the vacuum charge structure, while approximate continuous shift symmetry grades cross-harmonic mixing by explicit powers of ϵ . This is sufficient for the benchmark, but it does not excuse the calculation of the remaining finite operator tensors.

Third, the full decay chronology matters. A branching fraction that is exact at the R decay vertex is not automatically exact for the final temperatures when an earlier daughter deposits energy asymmetrically. The corrected branch is an output of the dynamical cascade, not a cosmetic retuning.

The resulting picture is:

vacuum action symmetry $\Rightarrow$ protected $p = 6$ vacuum harmonic,
 selector/state asymmetry $\Rightarrow$ temporary $p = 1$ focusing force,
 three-loop mixed matching $\Rightarrow$ tiny transient radiative correction,
 fermionic cascade $\Rightarrow$ controlled asymmetric population,
 full redshift history $\Rightarrow$ corrected $T_5/T_0 = 1/4$.

12. Next decisive calculation

The next target should combine two previously separate gaps:

RG-improved transient matching + non-Abelian nonequilibrium transport.

Concretely:

1. construct the complete local operator basis $e^{ipx} Q_{-p}$, $e^{ipx} Q_{-p} H^\dagger H$, $e^{ipx} Q_{-p} \bar{D} D$, and $e^{ipx} Q_{-p} G^2$, including evanescent and equation-of-motion operators as required;
2. calculate the one-loop and relevant two-loop $\Gamma_{AB}^{(r,s)}$ tensors;
3. run the hard coefficient from m_R through M and the electroweak scale, verifying matching-scale cancellation;
4. replace the BGK closure with explicit thermal collision kernels or a 2PI truncation;
5. scan Γ_N/Γ_R , thermal masses, M_D/T , and the selector decay profile to determine whether the corrected B_5 is stable or shifts appreciably;
6. feed the resulting temperature and perturbation transfer functions back into the cosmological vacuum-selection calculation.

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